

NBER WORKING PAPER SERIES

DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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Working Paper 25018

<http://www.nber.org/papers/w25018>

NATIONAL BUREAU OF ECONOMIC RESEARCH

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September 2018

I thank Michael Anderson, Martha Bailey, Marianne Bitler, Brantly Callaway, Kitt Carpenter, Eric Chyn, Bill Collins, John DiNardo, Andrew Dustan, Federico Gutierrez, Brian Kovak, Emily Lawler, Doug Miller, Sayeh Nikpay, Pedro Sant'Anna, Jesse Shapiro, Gary Solon, Isaac Sorkin, Sarah West, and seminar participants at the Southern Economics Association, ASHEcon 2018, the University of California, Davis, University of Kentucky, University of Memphis, University of North Carolina Charlotte, the University of Pennsylvania, and Vanderbilt University. All errors are my own. The views expressed herein are those of the author and do not necessarily reflect the views of the National Bureau of Economic Research.

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Difference-in-differences (DD) is both the most common and the oldest quasi-experimental research design, dating back to Snow’s (1855) analysis of a London cholera outbreak.¹ A DD estimate is the difference between the change in outcomes before and after a treatment (difference one) in a treatment versus control group (difference two): $(\bar{y}_{TREAT}^{POST} - \bar{y}_{TREAT}^{PRE}) - (\bar{y}_{CONTROL}^{POST} - \bar{y}_{CONTROL}^{PRE})$. That simple quantity also equals the estimated coefficient on the interaction of a treatment group dummy and a post-treatment period dummy in the following regression:

$$y_{it} = \gamma + \gamma_i TREAT_i + \gamma_t POST_t + \beta^{2x2} TREAT_i \times POST_t + u_{it} \quad (1)$$

The elegance of DD makes it clear which comparisons generate the estimate, what leads to bias, and how to test the design. The expression in terms of sample means connects the regression to potential outcomes and shows that, under a common trends assumption, a two-group/two-period (2x2) DD identifies the average treatment effect on the treated. All econometrics textbooks and survey articles describe this structure,² and recent methodological extensions build on it.³

Most DD applications diverge from this 2x2 set up though because treatments usually occur at different times.⁴ The processes that generate treatment variables naturally lead to variation in timing. Local governments change policy. Jurisdictions hand down legal rulings. Natural disasters strike across seasons. Firms lay off workers. In this case researchers estimate a regression with dummies for cross-sectional units (α_i) and time periods (α_t), and a treatment dummy (D_{it}):

$$y_{it} = \alpha_i + \alpha_t + \beta^{DD} D_{it} + e_{it} \quad (2)$$

¹ A search from 2012 forward of nber.org, for example, yields 430 results for “difference-in-differences”, 360 for “randomization” AND “experiment” AND “trial”, and 277 for “regression discontinuity” OR “regression kink”.

² This includes, but is not limited to, Angrist and Krueger (1999), Angrist and Pischke (2009), Heckman, Lalonde, and Smith (1999), Meyer (1995), Cameron and Trivedi (2005), Wooldridge (2010).

³ Inverse propensity score reweighting: Abadie (2005), synthetic control: Abadie, Diamond, and Hainmueller (2010), changes-in-changes: Athey and Imbens (2006), quantile treatment effects: Callaway, Li, and Oka (forthcoming).

⁴ Half of the 93 DD papers published in 2014/2015 in 5 general interest or field journals had variation in timing.

extend recent research on DD models with heterogeneous effects.⁷ Assuming equal counterfactual trends, Abraham and Sun (2018), Borusyak and Jaravel (2017), and de Chaisemartin and D’Haultfœuille (2018b) show that two-way fixed effects DD yields an average of treatment effects across all groups and times, some of which may have negative weights. My results show how these weights arise from differences in timing and thus treatment variances, facilitating a connection between models of treatment allocation and the interpretation of DD estimates.⁸ I also explain why the negative weights occur: when already-treated units act as controls, *changes* in their treatment effects over time get subtracted from the DD estimate. This negative weighting only arises when treatment effects vary over time, in which case it typically biases regression DD estimates away from the sign of the true treatment effect. This does not imply a failure of the underlying *design*, but it does caution against the use of a single-coefficient two-way fixed effects specification to summarize time-varying effects.

I also show that because regression DD uses group sizes and treatment variances to weight up simple estimates that each rely on common trends between two groups, its identifying assumption is a variance-weighted version of common trends between all groups. The extent to which a group’s differential trend biases the overall estimate equals the difference between how much weight it gets when it acts as the treatment group and how much weight it gets when it acts as the control group. When the earliest and/or latest treated units have low treatment variance, they can get *more* weight as controls than treatments. In designs without untreated units they always

⁷ Early research in this area made specific observations about stylized specifications such as models with no unit fixed effects (Bitler, Gelbach, and Hoynes 2003), or it provided simulation evidence (Meer and West 2013).

⁸ Related results on the weighting of heterogeneous treatment effects does not provide this intuition. Abraham and Sun (2018, p 9) describe the weights in a DD estimate with constant treatment effects as “residual[s] from predicting treatment status, $D_{i,t}$ with unit and time fixed effects.” de Chaisemartin and D’Haultfœuille (2018b, p 7) and Borusyak and Jaravel (2017) describe these same weights as coming from an auxiliary regression and Borusyak and Jaravel (2017, p 10-11) note that “a general characterization of [the weights] does not seem feasible.” Athey and Imbens (2018) also decompose the DD estimator and develop design-based inference methods for this setting. Strezhnev (2018) expresses $\hat{\beta}^{DD}$ as an unweighted average of DD-type terms across pairs of observations and periods.

post-treatment effect, which is closer to -5. My proposed balance test detects significantly higher per-capita income and male/female sex ratios in reform states, in contrast to joint tests of covariate balance across timing groups, which cannot reject the null of balance. Finally, my results show that much of the sensitivity across specifications comes from changes in weights, or a small number of 2x2 DD's, and need not indicate bias.

I. THE DIFFERENCE-IN-DIFFERENCES DECOMPOSITION THEOREM

When units experience treatment at different times, one cannot estimate equation (1) because the post-period dummy is not defined for control observations. Nearly all work that exploits variation in treatment timing uses the two-way fixed effects model in equation (2) (Cameron and Trivedi 2005 pg. 738). Researchers clearly recognize that differences in *when* units received treatment contribute to identification, but have not been able to describe how these comparisons are made.⁹

The simplest way to illustrate how treatment timing works is to consider a balanced panel dataset with T periods (t) and N cross-sectional units (i) that belong to either an untreated group, U ; an early treatment group, k , which receives a binary treatment at t_k^* ; and a late treatment group, ℓ , which receives the binary treatment at $t_\ell^* > t_k^*$. Figure 1 plots this structure.

Throughout the paper I use “group” or “timing group” to refer to collections of units either treated at the same time or not treated. I refer to units that do not receive treatment as “untreated” rather than “control” units because, while they obviously act as controls, treated units do, too. k will denote an earlier treated group and ℓ will denote a later treated group. Each group's sample share is n_k and the share of time it spends treated is $\bar{D}_k$. I use $\bar{y}_b^{POST(a)}$ to denote the sample mean of y_{it} for units in group b during group a 's post period, $[t_a^*, T]$. $(\bar{y}_b^{PRE(a)})$ is defined similarly.)

⁹ Angrist and Pischke (2015), for example, lay out the canonical DD model in terms of means, but discuss regression DD with timing in general terms only, noting that there is “more than one...experiment” in this setting.

The challenge in this setting has been to articulate how estimates of equation (2) compare the groups and times depicted in figure 1. We do, however, have clear intuition, for 2x2 designs in which one group's treatment status changes and another's does not. We could form several such designs, estimable by equation (1), in the three-group case. Figure 2 plots them.

Panels A and B show that with only one of the two treatment groups, an estimate from equation (2) reduces to the canonical case comparing a treated to an untreated group:

$$\hat{\beta}_{jU}^{2x2} \equiv \left(\bar{y}_j^{POST(j)} - \bar{y}_j^{PRE(j)} \right) - \left(\bar{y}_U^{POST(j)} - \bar{y}_U^{PRE(j)} \right), \quad j = k, \ell. \quad (3)$$

If instead there were no untreated units, the two way fixed effects estimator would be identified only by the differential treatment timing between groups k and ℓ . For this case, panels C and D plot two clear 2x2 DD's based on sub-periods when only one group's treatment status changes. Before t_ℓ^* , the early units act as the treatment group because their treatment status changes, and later units act as controls during their pre-period. We compare outcomes between the window when treatment status varies, $MID(k, \ell)$, and group k 's pre-period, $PRE(k)$:

$$\hat{\beta}_{k\ell}^{2x2,k} \equiv \left(\bar{y}_k^{MID(k,\ell)} - \bar{y}_k^{PRE(k)} \right) - \left(\bar{y}_\ell^{MID(k,\ell)} - \bar{y}_\ell^{PRE(k)} \right) \quad (4)$$

The opposite situation, shown in panel D, arises after t_k^* when the later group changes treatment status but the early group does not. Later units act as the treatment group, early units act as controls, and we compare average outcomes between the periods $POST(\ell)$ and $MID(k, \ell)$:

$$\hat{\beta}_{k\ell}^{2x2,\ell} \equiv \left(\bar{y}_\ell^{POST(\ell)} - \bar{y}_\ell^{MID(k,\ell)} \right) - \left(\bar{y}_k^{POST(\ell)} - \bar{y}_k^{MID(k,\ell)} \right) \quad (5)$$

The already-treated units in group k can serve as controls even though they are treated because treatment status does not change.

My central result is that any two-way fixed effects DD estimator is a weighted average of well-understood 2x2 DD estimators, like those plotted in figure 2. To see why, first assume a